

Signed Local-Response Extrapolation for Heterogeneous Quadratic Federated Optimization

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Abstract

Local optimization on heterogeneous clients changes the response seen by a federated server, even for strongly convex quadratic objectives. We study a common-restart construction that combines normalized local responses at dyadic step sizes with signed interpolation weights. A one-sided polynomial remainder controls the resulting operator despite the negative coefficients, yielding an explicit heterogeneity-bias bound and a finite-time mean-square estimate without requiring client Hessians to commute. A two-client scalar construction characterizes the attainable cancellation order within the class of fixed positive nodes and fixed weights, and quantifies the independent-noise cost of that cancellation. Under an alternative additive-noise model, shared innovations remove the weight-dependent part of the variance bound. At full interpolation order the realized update is exactly a minibatch-gradient update with the same innovations. These conclusions specialize established extrapolation and coupling principles to local quadratic responses within the fixed-node response class. Reproducible synthetic experiments exhibit the predicted bias powers and variance dependence, while an equal-gradient-call comparison favors direct minibatching in the tested model. The analysis isolates which benefits follow from response correction and which depend on access to a particular noise coupling.

1 Introduction

Federated optimization allows clients to perform several local computations between server synchronizations. This makes the relation between communication and local work central to algorithm design. The original federated averaging construction illustrates the practical appeal of this arrangement (McMahan et al., 2023); subsequent surveys and guidance emphasize that heterogeneous client data, participation patterns, and system costs must be distinguished when assessing its behavior (Kairouz et al., 2021; Wang et al., 2021). Even with full participation and identical amounts of local work, a local response can point toward a different fixed point from the minimizer of the global objective. Understanding this deterministic discrepancy is a useful preliminary step before considering sampling, partial participation, or nonlinear models.

For quadratic objectives, the discrepancy has a concrete algebraic form. Several local gradient steps multiply each client’s gradient by a polynomial in that client’s Hessian. Averaging these transformed gradients produces a surrogate global objective. The surrogate interpretation is established in the quadratic analysis of local updates (Charles & Konečný, 2020). It also explains why simply sending a more accurate solution of every local problem need not improve the global solution: a locally accurate response can still apply the wrong client-dependent transformation. Control-variate methods such as SCAFFOLD address client drift by a different mechanism (Karimireddy et al., 2021). We isolate what can be achieved by combining several local responses from the same starting point.

The construction uses r branches on each client. Every branch restarts at the current server iterate, performs K local steps, and returns its displacement divided by the branch step size and by K . The client combines these normalized responses using weights that interpolate their dependence on the local step size at zero. The server then applies a separate step size to the combined response. This arrangement is amenable to exact analysis because all branches in a round act on the same initial gradient. Its cost is explicit: rK gradient evaluations per client and one combined vector sent to the server. Sharing random innovations

between branches can reduce the number of distinct random draws, but it does not reduce this count of gradient evaluations.

Interpolation cancels low-order deterministic terms, yet that observation alone does not establish a useful optimization guarantee. The required weights have both signs. A triangle-inequality bound on their absolute values can obscure positivity of the effective Hessian and can yield an unnecessarily weak heterogeneity estimate. Moreover, small deterministic bias does not imply small stochastic error: independent branches pay a sum-of-squared-weights variance cost. The technical question is therefore whether the same local-response representation can simultaneously expose stability, the attainable bias order, and the dependence of noise on branch coupling.

We answer this question for deterministic, uniformly positive definite client Hessians and centered additive gradient noise. First, a signed interpolation remainder places every corrected local response operator in the interval $[(1 - \varepsilon)I, I]$. The interval survives aggregation without a common eigenbasis for different clients. Centering it at its midpoint and using stationarity at the global optimum yields the bias bound $\varepsilon\zeta/[2(1 - \varepsilon)\mu]$, where ζ measures client gradients at that optimum. An exact affine recursion then provides the finite-time stochastic estimate in Theorem 2. The result identifies a sufficient stability regime and a quantitative bias envelope.

Second, Theorem 3 describes a limitation of the particular correction class. For any fixed collection of distinct positive nodes and fixed weights, the first uncanceled moment determines the leading bias on an explicit two-client scalar problem. Fewer nodes than local steps cannot cancel all relevant moments. Nonnegative weights cannot cancel even the first moment. With independent scalar noise, the limiting response variance is proportional to the squared Euclidean norm of the weights. This identifies the attainable order within the fixed-node linear response class.

Third, Theorem 4 replaces branch independence with an explicit common-innovation model. A stronger local-step condition controls every noise propagation polynomial individually. The resulting variance proxy interpolates between the independent-branch factor and the ordinary minibatch factor. At interpolation order at least K , completely shared innovations give an exact samplewise minibatch-gradient response. The residual sampling noise remains; it is the amplification caused by the signed weights that disappears.

The relevant principles have substantial precedents. Higher-order extrapolation and Vandermonde cancellation are classical and have been developed for statistical learning (Bach, 2020). Richardson–Romberg correction has also been applied to the stationary bias of federated averaging (Mangold et al., 2025b), and common innovations are known to preserve limiting variance in stochastic extrapolation (Frikha & Huang, 2015). We develop the explicit, finite- K combination of these mechanisms for common-restart local quadratic responses, including a one-sided matrix bound, a scalar order witness, and a transfer-by-transfer coupling calculation. The distinction matters because a global-chain extrapolation and a within-round response extrapolation act on different random objects.

The numerical study follows the same separation. It measures fixed-point bias, response variance, and error against gradient calls as distinct quantities. Independently rotated client Hessians ensure that the examples do not rely on interclient commutation. Bias powers and the coupling trend agree with the formulas, but direct minibatching performs better at equal gradient-call budgets in the tested model. This outcome limits the operational conclusion: response correction offers a controlled way to study the bias–noise tradeoff, while practical efficiency requires additional structure beyond the stated oracle assumptions.

2 Related Work

Local updates and heterogeneous objectives. Federated averaging combines local computation and periodic model aggregation (McMahan et al., 2023). Analyses of local SGD quantify when this organization can reduce synchronization under particular stochastic and similarity assumptions (Stich, 2019). The broad federated-learning literature also distinguishes statistical heterogeneity from device availability, privacy, and systems constraints (Kairouz et al., 2021). Our full-participation quadratic model addresses only the first of these dimensions. Its local-work parameter K is common to all clients. FedNova instead ana-

lyzes unequal-work objective inconsistency: the normalization and effective objective in its Lemma 1 concern client-dependent amounts of local work (Wang et al., 2020).

Several algorithms explicitly correct or regularize local drift. FedProx introduces local proximal subproblems and analyzes inexact local work under a dissimilarity condition, including Definition 1, Assumption 1, and Theorem 4 (Li et al., 2020). SCAFFOLD uses control variates to correct the direction of local updates; its Section 3.2 explains this mechanism (Karimireddy et al., 2021). FedDyn uses dynamically changing local regularization, with convergence statements in its first two theorems (Acar et al., 2021). FedSplit instead uses an operator-splitting formulation: its Proposition 1 identifies a fixed-point issue and Theorem 1 studies an inexact proximal splitting method (Pathak & Wainwright, 2020). These methods use correction state, regularization, or proximal subproblems, whereas the present method uses fixed signed combinations of common-restart response polynomials.

Adaptive server methods change another part of the iteration. The FedOpt framework permits server-side adaptive updates and establishes guarantees under its own smoothness and stochastic assumptions (Reddi et al., 2021). ProxSkip obtains a communication benefit through randomized skipping and a control-variate mechanism; the dependence on the skipping probability in Theorem 3.6 is not represented by our deterministic full-participation schedule (Mishchenko et al., 2023). Unified local-SGD analyses based on shifted estimators cover additional algorithms under their stated estimator conditions (Gorbunov et al., 2020). These distinctions motivate a separate server step size α and explicit accounting of both gradient evaluations and synchronization rounds.

Quadratic surrogates and local drift. The closest algebraic starting point is the analysis of Charles & Konečný (2020). Their quadratic surrogate construction, including Equation 4, Theorem 1, and Lemma 1, expresses local updates as gradients of transformed quadratic objectives. We use the same underlying polynomial-response phenomenon. The signed combination requires an additional spectral argument because its coefficients have both signs. Our one-sided remainder provides a sufficient interval for that combination and makes the midpoint centering in the heterogeneity bound explicit. The polynomial representation itself is established prior work.

Wang et al. analyze why federated averaging can be effective through drift at the optimum (Wang et al., 2022). Their Lemma 2 develops a polynomial drift expression, while specific random-data and infinite-client arguments can produce vanishing averaged drift. We instead fix finitely many deterministic client Hessians and explicitly combine their responses to suppress drift. Global stationarity gives $\sum_i p_i g_i = 0$, while the transformed drift generally remains nonzero and is controlled by the response interval.

Bias–variance-reduced local SGD uses Hessian similarity and recursive corrections (Murata & Suzuki, 2021). Its correction state and assumptions differ from the common-restart branch construction. Refined analyses of local SGD and SCAFFOLD under Hessian similarity and weak convexity likewise concern a broader algorithmic comparison (Luo et al., 2025). An analysis of SCAFFOLD through its joint Markov chain gives stationary-distribution conclusions, including Theorem 4.2 (Mangold et al., 2025a). Our affine additive-noise model needs no evolving control-variate state and supports direct finite-time covariance calculations, but this simplification also removes important sources of difficulty in those methods.

Extrapolation and its variance. The Richardson principle cancels leading terms by evaluating a quantity at several scales. Bach’s treatment of extrapolation in statistical learning explicitly discusses higher-order Vandermonde cancellation in Section 3.1 (Bach, 2020). The weights and their moment identities are instances of this established interpolation construction. Mangold et al. apply Richardson–Romberg extrapolation to federated averaging; their Theorem 5 and Section 6 study bias correction using global chains at two step sizes (Mangold et al., 2025b). That result supplies a direct federated-learning precedent. Our branches restart at a common server iterate within every round and combine normalized displacements before the server update. The resulting finite polynomial has an exact full-order endpoint, whereas extrapolating separately evolving chains is a different operation. The common restart and finite-degree response are the specific objects of the present analysis.

The importance of coupling is also established. Frikha and Huang’s Remark 2.3 contrasts independent innovations, which incur a sum-of-squared-weights factor, with shared innovations that preserve limiting variance in multistep extrapolation (Frikha & Huang, 2015). Pagès gives corresponding coupled-diffusion results in Proposition 3.1 and analyzes covariance choices in Proposition 3.3 (Pagès, 2007). Our parameter θ selects one explicit additive mixture from the larger space of joint laws. The finite-step conclusion follows from controlling each local propagation polynomial under $\eta KL \leq 1$.

High-order extrapolation for Markovian linear stochastic approximation studies random matrices, a stable mean operator, and uniformly geometrically ergodic driving chains (Levin et al., 2025). Those assumptions support a substantially different stochastic analysis from deterministic Hessians with fresh additive innovations. The present exact samplewise identity relies on deterministic propagation and cannot be transferred to multiplicative or Markovian noise by replacing a variance symbol. Server extrapolation with line search, as studied by Geetika et al. (2024), is also distinct: extrapolating a server move does not automatically reproduce the polynomial cancellation of several local step sizes.

Limits and cost comparisons. Minibatch versus local-SGD analyses show why communication and stochastic-oracle budgets must be compared explicitly (Woodworth et al., 2022). The lower bounds of Patel et al. (2026) include a local-SGD result and a result for a broader class of zero-respecting distributed algorithms. Our Theorem 3 has neither of those scopes: it fixes positive nodes and linear combination weights, then exhibits a scalar witness within that class. The field guide’s separation of communication, computation, and statistical progress is particularly relevant to this comparison (Wang et al., 2021). Accordingly, the experiments charge every branch evaluation and include direct minibatching.

Finally, nonlinear local updates can exhibit iterate bias that is absent from our exact affine recursion. The definition and bounds of Glasgow et al. (2022) make this distinction concrete. Results on conservative iterated vector fields identify special nonlinear structures under which a surrogate interpretation can persist (Charles & Rush, 2021). The deterministic Hessian polynomial used in the present analysis is specific to the quadratic model. These boundaries explain both the tractability and the restricted application scope of the present results.

3 Quadratic Models and Local Responses

Objectives and heterogeneity. Let n, d, K, r be positive integers. Client weights satisfy $p_i > 0$ and $\sum_{i=1}^n p_i = 1$. Each client has the deterministic quadratic objective

$$f_i(x) = \frac{1}{2}x^\top H_i x - b_i^\top x, \quad \mu I \preceq H_i \preceq LI, \quad 0 < \mu \leq L, \quad (1)$$

where H_i is symmetric and $b_i \in \mathbb{R}^d$. We do not assume $H_i H_{i'} = H_{i'} H_i$. Write

$$\bar{H} = \sum_i p_i H_i, \quad x^* = \bar{H}^{-1} \sum_i p_i b_i, \quad g_i = H_i x^* - b_i, \quad \zeta^2 = \sum_i p_i \|g_i\|^2.$$

The weighted objective $f = \sum_i p_i f_i$ is strongly convex and $\sum_i p_i g_i = 0$. The quantity ζ is evaluated at its unique optimum. It permits substantial client disagreement; no uniform bound on all gradients over $\mathbb{R}^d$ is imposed. Norms are Euclidean for vectors and induced operator norms for matrices. Matrix inequalities use the Loewner order.

Common-restart response correction. Choose $\eta > 0$ with $\eta L \leq 1$. For $j = 0, \dots, r-1$, let

$$s_j = 2^{-j}, \quad w_j = \prod_{\substack{0 \leq \ell < r \\ \ell \neq j}} \frac{-s_\ell}{s_j - s_\ell}, \quad W_2 = \sum_{j=0}^{r-1} w_j^2. \quad (2)$$

An empty product equals one. These are the Lagrange weights for evaluation at zero, so $\sum_j w_j = 1$ and $\sum_j w_j s_j^\ell = 0$ for $1 \leq \ell < r$. For example, $r = 1$ gives $w_0 = 1$, while $r = 2$ gives $(-1, 2)$ at nodes $(1, 1/2)$.

At round t , all clients receive x_t . Each client executes the following equations for every branch:

$$\begin{aligned} y_{i,j,0,t} &= x_t, \\ y_{i,j,k+1,t} &= y_{i,j,k,t} - \eta s_j (H_i y_{i,j,k,t} - b_i + \xi_{i,j,k,t}), \end{aligned} \quad 0 \leq k < K, \quad (3)$$

$$\begin{aligned} d_{i,j,t} &= \frac{x_t - y_{i,j,K,t}}{\eta s_j K}, \quad u_{i,t} = \sum_{j=0}^{r-1} w_j d_{i,j,t}, \\ x_{t+1} &= x_t - \alpha \sum_i p_i u_{i,t}, \end{aligned} \quad 0 < \alpha \leq 1/L. \quad (4)$$

The server step α is distinct from the local scale η . The normalization in $d_{i,j,t}$ ensures that each response tends to the client gradient as η tends to zero. The $r = 1$ case is a normalized, server-damped local-averaging update. Identifying its parameters with a particular implementation of FedAvg requires accounting for this normalization.

One round requires rK gradient evaluations on each participating client. A client can combine the branches locally and send one vector of dimension d . The analysis assumes every client participates in every round and all clients use the same K . It neither charges a separate uplink for each branch nor treats branch computation as free. Branches may be parallelized, but parallel execution time is not analyzed.

Response polynomials and constants. Define

$$q_K(z) = \frac{1}{K} \sum_{k=0}^{K-1} (1-z)^k, \quad C_i = \sum_j w_j q_K(\eta s_j H_i), \quad S = \sum_i p_i H_i C_i. \quad (5)$$

The polynomial definition is valid also at $z = 0$ and avoids division by z . Set

$$\varepsilon = \begin{cases} \frac{(\eta L)^r}{r+1} \binom{K-1}{r} 2^{-r(r-1)/2}, & r < K, \\ 0, & r \geq K, \end{cases} \quad m = (1 - \varepsilon)\mu, \quad \rho = 1 - \alpha m. \quad (6)$$

We require $\varepsilon < 1$. Consequently $m > 0$ and $0 \leq \rho < 1$. This is a verifiable sufficient condition expressed in the selected local scale, local work, and interpolation order. It is not a statement that all larger scales are unstable.

Two alternative oracle laws. For Theorem 2, x_0 is deterministic and the vectors $\xi_{i,j,k,t}$ are mutually independent across all four indices, centered, and satisfy $\mathbb{E}\|\xi_{i,j,k,t}\|^2 \leq \sigma_i^2 < \infty$. The innovations are additive and independent of the current state. Their distributions need not be Gaussian or identical. Denote

$$V = \frac{W_2}{K} \sum_i p_i^2 \sigma_i^2. \quad (7)$$

Theorem 4 replaces this independence across branches with a different joint law. For $\theta \in [0, 1]$, let

$$\xi_{i,j,k,t} = \sqrt{\theta} e_{i,k,t} + \sqrt{1-\theta} e'_{i,j,k,t}, \quad (8)$$

where all e and e' vectors are jointly independent, centered, independent of deterministic x_0 , and each has second moment at most σ_i^2 on client i . The same $e_{i,k,t}$ is used across branches at the same local step. We then additionally require $\eta K L \leq 1$ and define

$$V_\theta = \frac{\theta + (1-\theta)W_2}{K} \sum_i p_i^2 \sigma_i^2. \quad (9)$$

This law specifies access to common additive innovations. Evaluating a general stochastic gradient twice on the same data example need not produce this law, because the difference between the stochastic and expected gradients may depend on the queried point.

The order obstruction uses an auxiliary scalar two-client construction and arbitrary fixed positive nodes. Its assumptions are stated separately in Theorem 3; the obstruction does not require arbitrary nodes to satisfy the dyadic stability condition for all η . It concerns the limit of sufficiently small positive η .

4 Stability and Heterogeneity Bias

Signed coefficients make direct stability arguments delicate: even when every uncorrected local response is positive definite, a general signed sum need not be. The dyadic weights in (2) have additional structure. Their error as an evaluation formula has a known sign over the entire spectrum allowed by (1). This sign is the central input to the following lemma.

Lemma 1 (One-sided response control). *Under the deterministic assumptions in Section 3, the operators in (5) satisfy*

$$(1 - \varepsilon)I \preceq C_i \preceq I, \quad mI \preceq S \preceq LI. \quad (10)$$

If $r \geq K$, then $C_i = I$. If $\eta KL \leq 1$ and $r < K$, then $\varepsilon \leq 2^{-r(r-1)/2}/(r+1)!$.

The lemma uses a scalar remainder before applying spectral calculus to each client separately. It therefore places no commutation requirement on different clients. Its proof is included in the complete proof of Theorem 2 in Appendix A.1.

Let $\tilde{b} = \sum_i p_i C_i b_i$ and define the mean fixed point $\bar{x} = S^{-1}\tilde{b}$. The existence of this inverse follows from the lemma, rather than being an additional assumption. Write

$$B = \frac{\varepsilon\zeta}{2m} = \frac{\varepsilon\zeta}{2(1-\varepsilon)\mu}.$$

The term B describes a deterministic fixed-point discrepancy. It is separate from the stochastic error around $\bar{x}$.

Theorem 2 (Bias, stability, and finite-time error). *Assume (1)–(6), $\varepsilon < 1$, deterministic x_0 , and the independent additive oracle law defining (7). The conditional mean update is $x \mapsto x - \alpha(Sx - \tilde{b})$, with unique fixed point $\bar{x}$. The spectral conclusions of Lemma 1 hold and*

$$\|\bar{x} - x^*\| \leq B. \quad (11)$$

For every integer $t \geq 0$,

$$\mathbb{E}\|x_t - \bar{x}\|^2 \leq \rho^{2t}\|x_0 - \bar{x}\|^2 + \alpha^2 V \frac{1 - \rho^{2t}}{1 - \rho^2}, \quad (12)$$

$$\mathbb{E}\|x_t - x^*\|^2 \leq 2\rho^{2t}\|x_0 - \bar{x}\|^2 + 2\alpha^2 V \frac{1 - \rho^{2t}}{1 - \rho^2} + 2B^2. \quad (13)$$

With zero noise, the stronger norm estimate

$$\|x_t - x^*\| \leq \rho^t\|x_0 - \bar{x}\| + B$$

holds. For $r \geq K$, the mean response is exactly the global gradient and $B = 0$. The complete derivation is in Appendix A.1.

Why the bias constant uses half the interval width. Stationarity gives $\sum_i p_i g_i = 0$, but the transformed sum $\sum_i p_i C_i g_i$ generally does not vanish. Subtracting a common scalar multiple of the identity leaves this transformed sum unchanged. The midpoint of the interval in (10) yields

$$\begin{aligned} S(\bar{x} - x^*) &= -\sum_i p_i C_i g_i \\ &= -\sum_i p_i (C_i - (1 - \varepsilon/2)I) g_i, \end{aligned} \quad (14)$$

$$\left\| \sum_i p_i (C_i - (1 - \varepsilon/2)I) g_i \right\| \leq \frac{\varepsilon}{2} \sum_i p_i \|g_i\| \leq \frac{\varepsilon\zeta}{2}.$$

Multiplication by S^{-1} gives (11). Centering at I would instead use the full width ε . The improvement follows from a deterministic operator interval and weighted stationarity; it does not require random cancellation of client gradients. The argument establishes this upper bound without asserting a matching worst-case constant.

Proof route. The first step differentiates q_K exactly. On $[0, 1]$, its r th derivative has sign $(-1)^r$ and absolute magnitude at most $r! \binom{K-1}{r} / (r+1)$. The Lagrange error at zero has the compensating sign, giving $0 \leq 1 - \sum_j w_j q_K(s_j z) \leq \varepsilon$ for $0 \leq z \leq \eta L$. The second step lifts this scalar interval into each client's eigenbasis. Within a client, H_i commutes with its polynomial C_i , so $H_i C_i$ is positive definite; summing these bounds does not multiply matrices from different clients. The third step unrolls the local recursions exactly, bounds the variance of their independent innovations, and iterates the conditional second-moment inequality. Appendix A.1 supplies the interpolation remainder, the local algebra, and the filtration calculation in full.

Interpretation of the regimes. For fixed K and $r < K$, ε is proportional to η^r . As η decreases, the fixed-point bias bound has the same order because m tends to μ . Increasing r reduces the deterministic remainder, but it changes the branch count and W_2 . The stochastic contribution in (12) therefore cannot be read from ε alone. For fixed α , its limiting upper bound is $\alpha^2 V / (1 - \rho^2)$; decreasing α changes both this term and the transient contraction.

The formula also separates several boundary cases. If all client gradients vanish at x^* , then $\zeta = 0$ and the fixed point is unbiased even when $\varepsilon > 0$. If all $\sigma_i = 0$, only deterministic contraction and bias remain. If $K = 1$, q_K is constant and there is no local-response bias to cancel. For $r \geq K$, interpolation is exact because the response polynomial has degree at most $K - 1$, but independent branch noise still need not be economical. The next result makes the latter distinction explicit.

5 Limits of Fixed-Node Correction

The preceding construction cancels powers of the local step size. To understand what that cancellation can achieve within its own class, fix m_0 distinct positive nodes $a_1, \dots, a_{m_0}$ and real weights $v_1, \dots, v_{m_0}$ with $\sum_j v_j = 1$. Both nodes and weights are independent of η . Define the moments $M_\ell = \sum_j v_j a_j^\ell$. The corrected scalar response at curvature h is

$$P_\eta(h) = \sum_j v_j q_K(\eta a_j h) = 1 + \sum_{\ell=1}^{K-1} \frac{\binom{K-1}{\ell}}{\ell+1} (-\eta h)^\ell M_\ell. \quad (15)$$

Thus the bias order is a question about finitely many moments, not only about the magnitude of a norm bound.

Theorem 3 (Fixed-node order and independent-noise cost). *Let $K \geq 2$ and use the fixed nodes and weights above. Consider two scalar clients with weights $1/2$, Hessians $H_1 = \mu < H_2 = L$, and linear coefficients $b_1 = -\zeta$, $b_2 = \zeta$, where $\zeta > 0$. Then $x^* = 0$. For sufficiently small positive η with $\eta(\max_j a_j)L \leq 1$, the mean fixed point $\bar{x}_\eta$ exists. If ℓ is the first nonzero moment among $M_1, \dots, M_{K-1}$, then*

$$\lim_{\eta \downarrow 0} \frac{|\bar{x}_\eta|}{\eta^\ell} = \frac{\zeta \binom{K-1}{\ell} |M_\ell| (L^\ell - \mu^\ell)}{(\ell+1)(L+\mu)}. \quad (16)$$

If $m_0 < K$, some moment among $M_1, \dots, M_{m_0}$ is nonzero, so universal bias order greater than m_0 is impossible within this fixed-node class. Cancellation through degree $m_0 - 1$ uniquely determines

$$v_j = \prod_{h \neq j} \frac{-a_h}{a_j - a_h}, \quad M_{m_0} = (-1)^{m_0+1} \prod_j a_j.$$

In particular, the dyadic construction has exact bias order r on this witness when $r < K$. Nonnegative weights obey $M_1 \geq \min_j a_j > 0$ and cannot cancel first order.

For a scalar client with curvature $h > 0$ and mutually independent centered additive noises of exact variance σ^2 , the corrected response noise Z_η satisfies

$$\lim_{\eta \downarrow 0} \text{Var}(Z_\eta) = \frac{\sigma^2}{K} \sum_j v_j^2. \quad (17)$$

For dyadic weights and $r \geq 2$, $W_2 > 1$. If $\sigma > 0$, the limiting variance ratio to a direct minibatch gradient using rK independent samples is $rW_2 > r$. If $\sigma = 0$, both variances vanish and no ratio is asserted. The complete proof is in Appendix A.2.

How the obstruction is obtained. The witness has nonzero client gradients of opposite signs at the global optimum. Its exact fixed point is

$$\bar{x}_\eta = -\frac{\zeta [P_\eta(\mu) - P_\eta(L)]}{\mu P_\eta(\mu) + L P_\eta(L)}.$$

The denominator converges to $\mu + L > 0$, while (15) identifies the first nonzero numerator coefficient. This gives the explicit nonzero limit in (16). To prove a moment must survive, apply the weighted evaluation functional to $\prod_j (u - a_j)$. That polynomial vanishes at every node but not at zero, contradicting exact evaluation of all powers through degree m_0 . Finally, the independent-noise calculation sums squared scalar propagation coefficients. As η tends to zero, each propagation coefficient tends to one, leaving the factor $\sum_j v_j^2/K$. These are separate deterministic and stochastic calculations, both completed in Appendix A.2.

Scope of the conclusion. Fixed nodes and weights make the moment equations independent of the observed trajectory. When $m_0 \geq K$, the response polynomial can instead be reproduced exactly. Within the regime $m_0 < K$, the scalar construction establishes a nonzero leading coefficient, while Theorem 2 controls the corresponding order over the stated matrix class. Together these statements identify the order of the dyadic construction and separate it from the question of optimal finite-step constants.

The variance comparison has an equally specific interpretation. With r branches, the algorithm spends rK calls on local trajectories. A direct minibatch at the server iterate can spend those same calls on rK independent gradients. Signed cancellation does not make these resources equivalent statistically: the branch combination attaches weights with squared sum exceeding one. This observation motivates studying dependence between branches, but changing that dependence changes the oracle law and must be analyzed explicitly.

6 Coupled Noise and Exact Recovery

A common innovation is propagated differently by different local step sizes. Controlling only the mean response C_i is therefore insufficient for its variance: one must control the transfer associated with every possible position of the innovation in the local trajectory. For a lag $k \in \{0, \dots, K-1\}$, define

$$R_{i,k} = \sum_j w_j (I - \eta s_j H_i)^k.$$

The extra condition $\eta KL \leq 1$ gives a useful one-sided bound on each such matrix.

Theorem 4 (Coupled transfer bound and samplewise endpoint). *Assume the deterministic model and parameters of Theorem 2, but replace its independent-branch oracle law by (8). Suppose also $\eta KL \leq 1$. For $k < r$, $R_{i,k} = I$. For $r \leq k < K$,*

$$(1 - \delta_k)I \preceq R_{i,k} \preceq I, \quad \delta_k = (\eta L)^r \binom{k}{r} 2^{-r(r-1)/2} \leq \frac{2^{-r(r-1)/2}}{r!} \leq 1. \quad (18)$$

The mean map, fixed point $\bar{x}$, and bias bound B are unchanged. The finite-time estimates (12) and (13) hold with V_θ in place of V . In particular,

$$\mathbb{E} \|x_t - x^*\|^2 \leq 2\rho^{2t} \|x_0 - \bar{x}\|^2 + 2\alpha^2 V_\theta \frac{1 - \rho^{2t}}{1 - \rho^2} + 2B^2. \quad (19)$$

If $\theta = 1$, the variance proxy is $K^{-1} \sum_i p_i^2 \sigma_i^2$. If also $r \geq K$, then, sample by sample,

$$u_{i,t} = H_i x_t - b_i + \frac{1}{K} \sum_{k=0}^{K-1} e_{i,k,t}. \quad (20)$$

Starting at the same x_0 , the entire server trajectory then equals that of the K -sample-per-client minibatch method using these same innovations.

For a scalar client of curvature $h > 0$, with all common and individual innovations of exact variance σ^2 , the response variance satisfies

$$\lim_{\eta \downarrow 0} \text{Var}(Z_\eta) = \frac{\sigma^2}{K} [\theta + (1 - \theta)W_2].$$

The full proof under this alternative joint law is in Appendix A.3.

Transfer-level covariance accounting. Unrolling (3) and combining branches expresses the client noise as

$$\begin{aligned} Z_{i,t} &= \frac{\sqrt{\theta}}{K} \sum_{k=0}^{K-1} R_{i,K-1-k} e_{i,k,t} \\ &\quad + \frac{\sqrt{1-\theta}}{K} \sum_{j,k} w_j (I - \eta s_j H_i)^{K-1-k} e'_{i,j,k,t}, \end{aligned} \quad (21)$$

$$\mathbb{E} \left\| \sum_i p_i Z_{i,t} \right\|^2 \leq \frac{\theta + (1-\theta)W_2}{K} \sum_i p_i^2 \sigma_i^2 = V_\theta.$$

Common innovations combine before their squared norm is bounded, which introduces $R_{i,k}$ in the first line. Individual innovations remain independent across branches and introduce W_2 . Treating the branch noises as independent under (8) would miss precisely this distinction.

The proof first interpolates the scalar polynomial $(1 - az)^k$ at the nodes s_j and evaluates it at $a = 0$. Its remainder is zero for $k < r$. Otherwise its sign is fixed and its size is bounded by $z^r \binom{k}{r} \prod_j s_j$. The condition $\eta KL \leq 1$ makes this bound at most one, so every common transfer has norm at most one. Next, independence of the underlying e and e' variables removes all cross terms in (21). Finally, conditioning on past innovations gives an affine second-moment recursion under this new law directly. The argument does not invoke branch independence from Theorem 2.

What exact recovery means. For $r \geq K$, all relevant deterministic response and noise-transfer polynomials have degree less than r . Interpolation therefore recovers their value at zero exactly. With $\theta = 1$, no individual-noise component remains, yielding (20). This is stronger than equality of expected gradients or equality of variance bounds: using the same realized innovations makes each update identical, and induction makes the trajectories identical.

The endpoint is also a resource boundary. It still computes rK local gradients per client, while the identical direct minibatch update uses K gradients at the common point and the same K innovations. The identity quantifies the additional local evaluations needed to reproduce this direct minibatch update. At lower orders, the method can retain a small deterministic bias while avoiding weight-dependent variance amplification under the prescribed coupling. Whether such a coupling can be implemented faithfully and economically for a particular data model is an additional question beyond the specified additive oracle.

7 Experiments: Bias, Noise, and Resource Costs

The experiments test three implications separately: the power of fixed-point bias in η , the dependence of response variance on θ , and the error obtained for a fixed number of gradient evaluations. They use a controlled quadratic model, so exact fixed points and covariances can be compared with sampled responses. The measurements illustrate the theorems under their assumptions; they do not test a deployed federated system.

Configuration and reproducibility. We use $n = 5$ clients, dimension $d = 4$, uniform weights, $K = 8$, $\mu = 0.2$, $L = 1$, $\zeta = 1$, $\sigma_i = 0.25$, and server step $\alpha = 0.5$. Each client's Hessian is an independently rotated matrix with eigenvalues 0.2, $0.2 + 0.8/3$, $0.2 + 1.6/3$, 1. Client linear terms are centered across clients and scaled to make $\zeta = 1$, so $x^* = 0$. The smallest measured Frobenius norm of $H_1 H_2 - H_2 H_1$ across the 20 problems is 0.172, confirming that the examples are not simultaneously diagonal constructions.

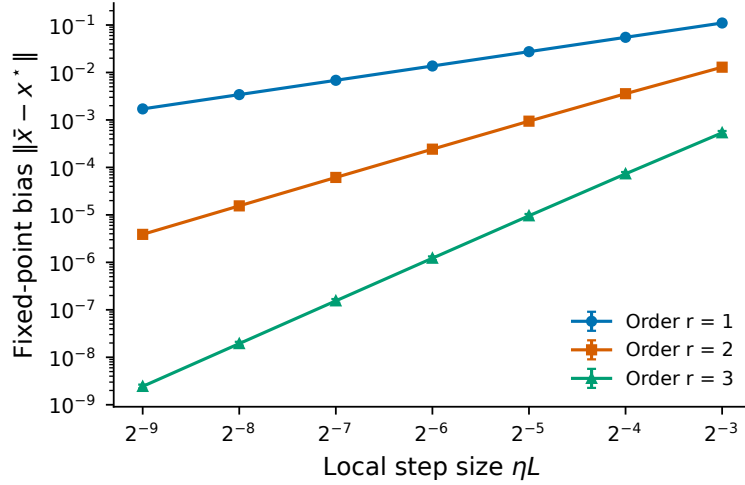


Figure 1: Fixed-point bias versus local step size in the synthetic heterogeneous quadratic model (five clients, dimension four, $K = 8$, $\mu = 0.2$, $L = 1$, $\zeta = 1$). Markers show means over 20 independently generated problems; error bars are one standard error across problem seeds. Slopes fitted to the four smallest steps are 1.001, 1.988 and 2.991 for $r = 1, 2, 3$. Deterministic fixed points are computed for each problem; their scaling illustrates the predicted cancellation orders.

Problem seeds are 1000, ..., 1019 and sampling seeds are 20000, ..., 20019. Gaussian innovations have covariance $\sigma^2 I/d$, so their expected squared norm is σ^2 , consistent with the theorem’s vector convention. Each stochastic configuration uses 256 responses or trajectories per problem seed. Error bars report one standard error across the 20 problem-level measurements. They quantify variation in this numerical study and are not simultaneous confidence bounds for the theorem.

The accompanying code computes response operators, fixed points, and exact covariance matrices in double precision and saves all per-seed results. The response-variance experiment explicitly executes the K local steps in every branch. For trajectory comparisons, we sample the mathematically identical affine Gaussian transition obtained from those operators and charge the nominal local gradient calls. This avoids confusing the execution time of matrix simulation with the runtime of a federated implementation.

Bias scaling. For $r = 1, 2, 3$, we evaluate seven scales ηL from $1/512$ through $1/8$. Figure 1 shows the resulting fixed-point bias. Fits to the four smallest scales give slopes 1.001, 1.988, and 2.991, respectively. The largest observed ratio of the exact bias to the bound B is 0.212. The calculations are consistent with the upper-bound order and with the nonzero-order witness, while their slack gives no evidence that the constant in B is attained by these matrix examples.

Coupled response variance. At $r = 3$ and $\eta = 1/8$, we vary θ over $0, 1/4, 1/2, 3/4, 1$. Figure 2 compares empirical centered second moments with exact covariance traces and the theorem proxy, normalized by $\sigma^2/(nK)$. At $\theta = 0$, the normalized empirical value is 9.411 with standard error 0.092; at $\theta = 1$, it is 1.000 with standard error 0.007. The corresponding exact finite-step values are 9.479 and 0.999. The independent proxy uses $W_2 = 101/9$, whereas the finite-step filters reduce the exact variance below that proxy. The common-noise endpoint is close to one at this lower interpolation order, but the theorem’s exact samplewise identity requires $r \geq K$ as well.

Gradient calls and communication rounds. We compare normalized local averaging ($r = 1$, 8 calls per client per round), independent correction ($r = 2$, 16 calls), common-noise correction ($r = 2$, 16 calls), and direct minibatching with 16 samples per client per round. All methods use $\alpha = 0.5$, start at the all-ones vector, and run for 120 rounds. Local-response methods use $\eta = 1/8$. Figure 3 uses gradient calls on the

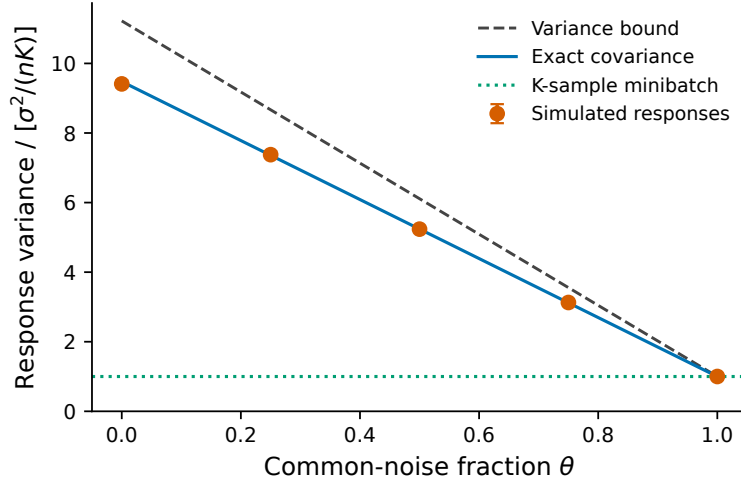


Figure 2: Coupling-dependent aggregate response variance for $r = 3$, $K = 8$ and $\eta L = 1/8$ in the synthetic quadratic additive Gaussian model. The blue line is the exact covariance trace, the dashed line is the theorem variance proxy, and orange markers are measured centered response second moments from 256 independently sampled local responses per problem seed, averaged over 20 problems; bars show one standard error across those problem means. Variance is normalized by $\sigma^2/(nK)$, the dotted K -sample minibatch reference, with $n = 5$ and $\sigma = 0.25$. Common-noise coupling removes the weight-dependent amplification in the bound while retaining nonzero noise. This is a response variance comparison at fixed K and r .

horizontal axis; one round corresponds to one combined client uplink for every method. The local-averaging curve ends earlier on that axis because it performs fewer calls per round.

At a common budget of 768 calls per client, the mean squared errors are 1.426×10^{-2} for local averaging, 3.314×10^{-3} for independent correction, 9.827×10^{-4} for common-noise correction, and 4.019×10^{-4} for minibatching. These values correspond to 96 local-averaging rounds and 48 rounds for the other methods. At 120 communication rounds, the three 16-call methods attain errors 3.246×10^{-3} , 9.795×10^{-4} , and 3.970×10^{-4} , respectively. Thus coupling helps the corrected method in this model, while direct minibatching remains better at both reported equal-resource comparisons. No wall-clock or communication-latency gain is measured.

Boundary and stress checks. With $K = r = 3$ and $\theta = 1$, the largest entrywise discrepancy from identity among the response and noise-transfer matrices is below 10^{-15} . This checks the polynomial endpoint numerically, in addition to its samplewise proof. Across the trajectory configurations, the largest ratio of exact mean squared error to the applicable theorem bound is 0.504 or less. We also record spectra at local scales $1/8, 1/4, 1/2, 1, 1.5$, including scales outside $\eta KL \leq 1$ and outside the stated local-step regime. These records distinguish certified conditions from observed behavior. Stability observed outside a sufficient condition does not extend the theorem, and no result here is inferred from those out-of-regime cases.

8 Limitations

The conclusions depend on deterministic quadratic Hessians, full client participation, and additive state-independent innovations. General empirical-risk gradients can have state-dependent noise and random Hessians; using the same example in different branches then need not implement (8). The common-noise theorem therefore establishes a property of the specified joint oracle, not a guarantee for arbitrary data reuse. The experiments use synthetic objectives and injected Gaussian innovations. They provide no direct observation of this coupling law in a real federated training system.

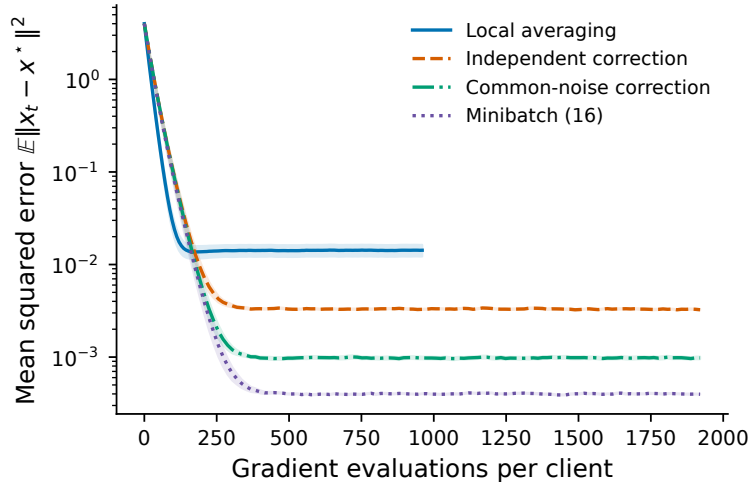


Figure 3: Mean squared error versus cumulative nominal gradient evaluations per client in the synthetic quadratic model ($K = 8$, $\eta L = 1/8$, $\alpha = 0.5$, $\sigma = 0.25$). Lines average 256 trajectories on each of 20 independently generated problems; shaded bands show one standard error across problem means. Local averaging uses eight nominal calls per round, while the two $r = 2$ corrections and the 16-sample minibatch baseline use sixteen; every method runs for 120 rounds, explaining the shorter local-averaging curve. Trajectories are sampled from the exact affine Gaussian update law, so the horizontal axis records oracle charges rather than measured runtime. Common-noise correction lowers error relative to independent correction in this setting, while the equal-call minibatch attains a lower final error.

The condition $\varepsilon < 1$ certifies the signed mean response, and $\eta KL \leq 1$ additionally certifies each common-noise transfer. Both are sufficient conditions. The analysis does not characterize the largest stable step or choose an optimal order after accounting for computation. Likewise, the order obstruction covers fixed nodes and weights, and the variance comparison concerns the stated scalar oracle. It cannot be used as a lower bound against adaptive federated algorithms.

Computational accounting is consequential: correction uses rK calls per client even when innovations are shared. The synthetic equal-call comparison favors direct minibatching, and the full-order shared-noise endpoint reproduces a minibatch update with additional local evaluations. Practical benefits would require a setting where local response access, parallelism, communication, or oracle structure changes this tradeoff. Such a benefit is not established by the present experiments.

9 Conclusion

Common-restart signed extrapolation makes three aspects of heterogeneous quadratic local optimization explicit. A one-sided response remainder yields stability and a centered heterogeneity-bias bound; fixed-node moments constrain the attainable order; and noise-transfer polynomials determine how branch coupling changes variance. At full order with shared additive innovations, the update equals a minibatch-gradient update sample by sample. The numerical results support the predicted powers and coupling dependence, while direct minibatching attains lower error at equal call budgets in the tested model. Extending the argument to state-dependent stochastic gradients requires controlling the interaction between evolving branch states and their joint noise, which is absent from the affine model.

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Appendix

A Proofs of the Main Results

This appendix proves the three main statements under their respective oracle laws. The independent-innovation argument and the coupled-innovation argument use different filtrations; their deterministic response algebra is common. All sums over local lags are finite, so the variance calculations below require only the stated finite second moments, with no exchange of an infinite series and an expectation.

A.1 Proof of Theorem 2

Proof. We first establish Lemma 1, then derive the fixed point and the finite-time estimates.

Interpolation identities and the sign of the remainder. For distinct nodes $s_0, \dots, s_{r-1}$, define

$$L_j(a) = \prod_{\ell \neq j} \frac{a - s_\ell}{s_j - s_\ell}.$$

The polynomials L_j have degree at most $r - 1$ and take value one at node j and zero at the others. For every polynomial P of degree at most $r - 1$, the difference $P(a) - \sum_j P(s_j) L_j(a)$ has r distinct roots and degree at most $r - 1$, hence is identically zero. Evaluation at zero gives $w_j = L_j(0)$ and

$$\sum_j w_j = 1, \quad \sum_j w_j s_j^\ell = 0 \quad (1 \leq \ell < r). \quad (22)$$

For completeness, the scalar interpolation remainder at zero follows directly from Rolle's theorem. Let F be r times continuously differentiable on $[0, 1]$, let P interpolate it at the nodes, and put $c = [F(0) - P(0)] / \prod_j (-s_j)$. The function $F(a) - P(a) - c \prod_j (a - s_j)$ vanishes at zero and at all r nodes. Repeated Rolle's theorem gives a point $a_* \in (0, 1)$ where its r th derivative is zero. Since $P^{(r)} = 0$ and the monic product has r th derivative $r!$,

$$F(0) - \sum_j w_j F(s_j) = \frac{F^{(r)}(a_*)}{r!} \prod_j (-s_j).$$

This argument needs no sign condition on the weights.

Apply it to $F(a) = q_K(az)$ for $0 < z \leq \eta L \leq 1$. When $r < K$, differentiating the finite polynomial gives

$$\frac{(-1)^r}{r!} q_K^{(r)}(u) = \frac{1}{K} \sum_{k=r}^{K-1} \binom{k}{r} (1-u)^{k-r}, \quad 0 \leq u \leq 1. \quad (23)$$

Every term is nonnegative and at most its binomial coefficient. Pascal's identity telescopes to

$$\sum_{k=r}^{K-1} \binom{k}{r} = \sum_{k=r}^{K-1} \left[\binom{k+1}{r+1} - \binom{k}{r+1} \right] = \binom{K}{r+1}.$$

Consequently the right side of (23) lies between zero and $\binom{K}{r+1}/K = \binom{K-1}{r}/(r+1)$. The factor z^r from differentiating F and the two signs $(-1)^r$ in its derivative and in the node product give

$$0 \leq 1 - \sum_j w_j q_K(s_j z) \leq z^r \frac{\binom{K-1}{r}}{r+1} \prod_j s_j. \quad (24)$$

At $z = 0$, the same assertion follows from $\sum_j w_j = 1$. Dyadic nodes have $\prod_j s_j = 2^{-r(r-1)/2}$. If $r \geq K$, the polynomial $a \mapsto q_K(az)$ has degree at most $K - 1 < r$, so (22) gives exact equality to one rather than requiring a remainder.

Spectral lifting and aggregate stability. Diagonalize a single client Hessian as $H_i = U_i \Lambda_i U_i^\top$, with U_i orthogonal and every eigenvalue in $[\mu, L]$. A polynomial in H_i has the same eigenvectors. Applying (24) at $z = \eta\lambda$ for each eigenvalue yields

$$(1 - \varepsilon)I \preceq C_i \preceq I.$$

Within this client, $H_i C_i = C_i H_i$ is symmetric and its eigenvalues are products $\lambda c_i(\lambda)$ with $1 - \varepsilon \leq c_i(\lambda) \leq 1$. It follows that

$$(1 - \varepsilon)H_i \preceq H_i C_i \preceq H_i.$$

Summing these inequalities with positive weights gives

$$(1 - \varepsilon)\bar{H} \preceq S \preceq \bar{H}, \quad mI \preceq S \preceq LI.$$

No product involving distinct client Hessians occurs, so their lack of commutation is irrelevant to this step. Since $\varepsilon < 1$, $m > 0$, S is invertible and $\|S^{-1}\| \leq 1/m$. Furthermore, for $M = I - \alpha S$, every eigenvalue lies in $[1 - \alpha L, 1 - \alpha m] \subseteq [0, \rho]$. Thus $\|M\| \leq \rho < 1$.

If $\eta KL \leq 1$ and $r < K$, then

$$(\eta L)^r \binom{K-1}{r} \leq K^{-r} \frac{(K-1)(K-2)\cdots(K-r)}{r!} \leq \frac{1}{r!}.$$

Substitution into (6) proves the additional bound $\varepsilon \leq 2^{-r(r-1)/2}/(r+1)!$ and completes the proof of Lemma 1.

Exact local recursion. Fix i, j, t , write $a = \eta s_j > 0$ and $Q = I - aH_i$, and temporarily suppress indices. The recursion $y_{k+1} = Qy_k + ab_i - a\xi_k$ with $y_0 = x_t$ gives by induction

$$y_K = Q^K x_t + a \sum_{k=0}^{K-1} Q^{K-1-k} b_i - a \sum_{k=0}^{K-1} Q^{K-1-k} \xi_k.$$

The polynomial identity $I - Q^K = (I - Q) \sum_{\ell=0}^{K-1} Q^\ell$ and $I - Q = aH_i$ imply

$$d_{i,j,t} = q_K(\eta s_j H_i)(H_i x_t - b_i) + \frac{1}{K} \sum_{k=0}^{K-1} (I - \eta s_j H_i)^{K-1-k} \xi_{i,j,k,t}. \quad (25)$$

Division by aK is legitimate because η, s_j, K are positive. This formula also holds for $K = 1$, when the propagation power is zero. Summing with weights w_j and then p_i gives

$$x_{t+1} = x_t - \alpha(Sx_t - \tilde{b} + Z_t), \quad Z_t = \frac{1}{K} \sum_{i,j,k} p_i w_j (I - \eta s_j H_i)^{K-1-k} \xi_{i,j,k,t}. \quad (26)$$

The matrices and coefficients are deterministic.

Fixed-point discrepancy. The affine mean map has fixed point $\bar{x} = S^{-1}\tilde{b}$; uniqueness follows from invertibility. At x^* , $Sx^* - \tilde{b} = \sum_i p_i C_i (H_i x^* - b_i) = \sum_i p_i C_i g_i$. Using $\sum_i p_i g_i = 0$ and the spectral interval already proved,

$$\begin{aligned} \|S(\bar{x} - x^*)\| &= \left\| \sum_i p_i (C_i - (1 - \varepsilon/2)I) g_i \right\| \\ &\leq \sum_i p_i \|C_i - (1 - \varepsilon/2)I\| \|g_i\| \\ &\leq \frac{\varepsilon}{2} \sum_i p_i \|g_i\| \leq \frac{\varepsilon}{2} \left(\sum_i p_i \|g_i\|^2 \right)^{1/2}. \end{aligned}$$

The last inequality is Cauchy–Schwarz with weights p_i and $\sum_i p_i = 1$. Applying $\|S^{-1}\| \leq 1/m$ proves $\|\bar{x} - x^*\| \leq B$. This explains every factor in the denominator and the midpoint factor $1/2$.

Conditional moments and finite-time bound. Let $\mathcal{F}_t$ be the sigma field generated by all innovations in rounds strictly before t together with deterministic x_0 . Induction in (26) shows that x_t is $\mathcal{F}_t$ -measurable and has finite second moment: the initial second moment is finite, each update is affine, and the current finite sum of innovations has finite second moment. By mutual independence and centering, $\mathbb{E}[Z_t | \mathcal{F}_t] = 0$. Each propagation matrix has norm at most one because its base has eigenvalues in $[0, 1]$. Cross terms between different innovations vanish, conditionally as well as unconditionally. Hence

$$\mathbb{E}[\|Z_t\|^2 | \mathcal{F}_t] \leq \frac{1}{K^2} \sum_{i,j,k} p_i^2 w_j^2 \sigma_i^2 = \frac{W_2}{K} \sum_i p_i^2 \sigma_i^2 = V.$$

Centering (26) at $\bar{x}$ yields $x_{t+1} - \bar{x} = M(x_t - \bar{x}) - \alpha Z_t$. Its mixed inner product has conditional expectation zero. Therefore

$$\mathbb{E}[\|x_{t+1} - \bar{x}\|^2 | \mathcal{F}_t] \leq \rho^2 \|x_t - \bar{x}\|^2 + \alpha^2 V.$$

Taking expectations and iterating gives

$$\mathbb{E}\|x_t - \bar{x}\|^2 \leq \rho^{2t} \|x_0 - \bar{x}\|^2 + \alpha^2 V \sum_{s=0}^{t-1} \rho^{2s}.$$

Since $0 \leq \rho < 1$, the sum is $(1 - \rho^{2t})/(1 - \rho^2)$. This is (12). The inequality $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$, applied to $x_t - x^* = (x_t - \bar{x}) + (\bar{x} - x^*)$, gives (13). When all noise moments are zero, the innovations vanish almost surely; deterministic iteration gives $\|x_t - \bar{x}\| \leq \rho^t \|x_0 - \bar{x}\|$, and the triangle inequality gives the stated norm estimate.

Boundary checks and conclusion. At $t = 0$, the finite sum is empty and equals zero; we use $\rho^0 = 1$, also when $\rho = 0$. Since $\rho < 1$, no denominator in the geometric expression vanishes. If $r \geq K$, interpolation gives $C_i = I$, so $S = \bar{H}$, $\tilde{b} = \sum_i p_i b_i$, and $\bar{x} = x^*$ exactly. The case $K = 1$ is included. If $\zeta = 0$, the bias bound is zero; if the noise moments vanish, the stochastic term is zero. If $\mu = L$, the spectral argument remains valid, with all Hessians equal to μI . Thus no strict inequality between the curvature bounds is needed for Theorem 2. The spectral interval, fixed-point statement, bias estimate, and all finite-time conclusions have now been proved under exactly its independent additive oracle law. $\square$

A.2 Proof of Theorem 3

Proof. We prove the polynomial expansion, derive the scalar witness, establish the node obstruction, and then calculate the independent-noise variance. The scalar noise calculation is separate from the deterministic witness.

Exact expansion and its first surviving term. The binomial theorem gives

$$(1 - z)^k = \sum_{\ell=0}^k \binom{k}{\ell} (-z)^\ell.$$

Summing over k in q_K , collecting equal powers, and using the telescoping binomial identity proved above yields

$$q_K(z) = \sum_{\ell=0}^{K-1} \left(\frac{1}{K} \sum_{k=\ell}^{K-1} \binom{k}{\ell} \right) (-z)^\ell = \sum_{\ell=0}^{K-1} \beta_\ell (-z)^\ell, \quad \beta_\ell = \frac{\binom{K-1}{\ell}}{\ell+1}.$$

In particular $\beta_0 = 1$. Since the node weights sum to one,

$$P_\eta(h) = \sum_j v_j q_K(\eta a_j h) = 1 + \sum_{\ell=1}^{K-1} \beta_\ell (-\eta h)^\ell M_\ell,$$

which proves (15). It is an exact finite polynomial identity, not an asymptotic approximation.

For the two scalar clients, the global linear term is zero, so $x^* = 0$. Their gradients there are ζ and $-\zeta$, hence the heterogeneity norm is indeed ζ . Their mean response is

$$\frac{1}{2}\{\mu P_\eta(\mu) + L P_\eta(L)\}x + \frac{\zeta}{2}\{P_\eta(\mu) - P_\eta(L)\}.$$

At $\eta = 0$ its slope is $(\mu + L)/2 > 0$. Continuity of this polynomial slope gives a positive neighborhood in which it is nonzero; intersecting that neighborhood with $0 < \eta \leq 1/[(\max_j a_j)L]$ is possible because the nodes are fixed and positive. The mean fixed point in that neighborhood is

$$\bar{x}_\eta = -\frac{\zeta [P_\eta(\mu) - P_\eta(L)]}{\mu P_\eta(\mu) + L P_\eta(L)}.$$

If M_ℓ is the first nonzero moment among degrees 1 through $K - 1$, division of the numerator by η^ℓ gives the limit $\zeta(-1)^{\ell+1}\beta_\ell M_\ell(\mu^\ell - L^\ell)$, while the denominator tends to $\mu + L$. Its limiting absolute value is $\zeta\beta_\ell|M_\ell|(L^\ell - \mu^\ell)/(\mu + L)$. All factors are strictly positive except for signs already removed: $\zeta > 0$, $L > \mu > 0$, $\beta_\ell > 0$, and $M_\ell \neq 0$. Thus the limit is nonzero and equals (16).

A moment must survive. Suppose $m_0 < K$ and, toward a contradiction, $M_1 = \dots = M_{m_0} = 0$. Set

$$Q(u) = \prod_{j=1}^{m_0} (u - a_j) = u^{m_0} + c_{m_0-1}u^{m_0-1} + \dots + c_0.$$

The constant term is $c_0 = (-1)^{m_0} \prod_j a_j \neq 0$. Evaluating at the nodes and weighting gives $\sum_j v_j Q(a_j) = 0$, because every evaluation is zero. On the other hand, expansion in powers and the hypothesized moment cancellations give

$$\sum_j v_j Q(a_j) = M_{m_0} + c_{m_0-1}M_{m_0-1} + \dots + c_1M_1 + c_0 \sum_j v_j = c_0 \neq 0.$$

This contradiction proves that some moment of degree at most m_0 survives. That degree is present in the response polynomial because $m_0 < K$. The witness calculation then rules out an order greater than m_0 uniformly over the stated class of heterogeneous quadratics.

Now suppose only $M_1 = \dots = M_{m_0-1} = 0$. The weighted evaluation functional agrees with evaluation at zero on every polynomial of degree at most $m_0 - 1$. Applying it to the Lagrange basis polynomial $\prod_{h \neq j} (u - a_h)/(a_j - a_h)$ gives $v_j = \prod_{h \neq j} (-a_h)/(a_j - a_h)$. This also proves uniqueness, without an unproved rank assertion. Conversely, these Lagrange weights reproduce every such polynomial by the root argument in (22), so they do provide the stated cancellations.

Applying the functional to Q again, with only the lower moments zero, gives $0 = M_{m_0} + c_0$, hence $M_{m_0} = (-1)^{m_0+1} \prod_j a_j$. This is nonzero. Dyadic weights have precisely these interpolation cancellations and a nonzero moment at degree r ; when $r < K$, the witness therefore has exact order r . If all $v_j \geq 0$, positivity and normalization instead give $M_1 = \sum_j v_j a_j \geq (\min_j a_j) \sum_j v_j = \min_j a_j > 0$. Thus first-order cancellation needs a negative weight within the fixed positive-node class.

Independent scalar variance. For one scalar client, the noise part of the normalized response at node a_j is $K^{-1} \sum_{k=0}^{K-1} (1 - \eta a_j h)^k \xi_{j,K-1-k}$, by the exact local unrolling. Combining nodes produces

$$Z_\eta = \frac{1}{K} \sum_{j=1}^{m_0} \sum_{k=0}^{K-1} v_j (1 - \eta a_j h)^k \xi_{j,K-1-k}.$$

The noises are centered, mutually independent, and have exact variance σ^2 . Each cross covariance therefore vanishes, giving the exact expression

$$\text{Var}(Z_\eta) = \frac{\sigma^2}{K^2} \sum_{j=1}^{m_0} v_j^2 \sum_{k=0}^{K-1} (1 - \eta a_j h)^{2k}.$$

The number of terms is fixed and finite. Taking $\eta \downarrow 0$ sends every propagation factor to one and yields $\sigma^2 \sum_j v_j^2/K$, proving (17). This passage uses continuity of a finite polynomial, so no uniform-integrability premise is missing.

For dyadic nodes, examine the coefficient of the smallest node s_{r-1} . Every factor in its defining product is positive and

$$w_{r-1} = \prod_{j=0}^{r-2} \frac{s_j}{s_j - s_{r-1}} = \prod_{k=1}^{r-1} (1 - 2^{-k})^{-1}.$$

For $r \geq 2$, each factor exceeds one, so $w_{r-1} > 1$ and $W_2 \geq w_{r-1}^2 > 1$. A direct gradient evaluated at a fixed scalar point and averaged over rK independent noises has exact variance $\sigma^2/(rK)$. When $\sigma > 0$, division is legitimate and the limiting ratio is $(\sigma^2 W_2/K)/(\sigma^2/(rK)) = rW_2 > r$.

Boundary checks and conclusion. The witness requires $L > \mu$ and $\zeta > 0$; setting either curvature difference or heterogeneity to zero would remove the nonzero leading coefficient and is not part of that assertion. The node obstruction requires $m_0 < K$. If $m_0 \geq K$, reproduction of a polynomial of degree $K - 1$ is possible, so the contradiction is not applied. Similarly $K = 1$ has no nonconstant response moment and is excluded from the obstruction statement. For $r = 1$, $W_2 = 1$, so the strict weight-cost conclusion is correctly restricted to $r \geq 2$. If $\sigma = 0$, both compared noises vanish almost surely, and no variance ratio is taken. These checks establish the complete fixed-node theorem with its stated class restrictions and exact independent-noise limit. $\square$

A.3 Proof of Theorem 4

Proof. This proof uses the alternative joint oracle (8). In particular, the noises indexed by different branches are generally dependent. We derive the stochastic recursion from the independent underlying e and e' innovations instead of importing the independent-branch variance calculation.

Polynomial control at every lag. Fix a client eigenvalue $\lambda \in [\mu, L]$, write $z = \eta\lambda$, and consider $F_k(a) = (1 - az)^k$ for a lag $0 \leq k < K$. If $k < r$, the polynomial has degree less than r , so interpolation at s_j gives $\sum_j w_j (1 - s_j z)^k = F_k(0) = 1$. If $k \geq r$, its r th derivative is

$$\frac{F_k^{(r)}(a)}{r!} = (-z)^r \binom{k}{r} (1 - az)^{k-r}.$$

The additional step condition implies $0 \leq z \leq \eta L \leq 1/K \leq 1$. For $a \in [0, 1]$, the last power is therefore nonnegative and at most one. Applying the remainder argument from the first proof, with its sign retained, gives

$$0 \leq 1 - \sum_j w_j (1 - s_j z)^k \leq z^r \binom{k}{r} \prod_j s_j \leq (\eta L)^r \binom{k}{r} 2^{-r(r-1)/2} = \delta_k.$$

For $k \geq r$, the factorial bound and $k < K$ give

$$(\eta L)^r \binom{k}{r} \leq K^{-r} \frac{k(k-1) \cdots (k-r+1)}{r!} \leq \frac{1}{r!}.$$

Thus $\delta_k \leq 2^{-r(r-1)/2}/r! \leq 1$. Spectral calculus in the eigenbasis of H_i now proves (18), and also proves $R_{i,k} = I$ for $k < r$. In either case $\|R_{i,k}\| \leq 1$. This last conclusion concerns every lag, which is stronger than a bound on their average.

Deterministic mean and fixed point under the coupled law. The exact algebraic unrolling (25) did not use independence. Substituting (8) into that identity gives

$$u_{i,t} = C_i(H_i x_t - b_i) + Z_{i,t},$$

where

$$Z_{i,t} = \frac{\sqrt{\theta}}{K} \sum_{k=0}^{K-1} R_{i,K-1-k} e_{i,k,t} + \frac{\sqrt{1-\theta}}{K} \sum_{j=0}^{r-1} \sum_{k=0}^{K-1} w_j (I - \eta s_j H_i)^{K-1-k} e'_{i,j,k,t}.$$

The deterministic part is unchanged because each underlying innovation is centered and every filter is deterministic. The scalar proof of (24) applies to C_i without any noise assumption. Accordingly, $(1-\varepsilon)I \preceq C_i \preceq I$ and $mI \preceq S = \sum_i p_i H_i C_i \preceq LI$. Writing $\tilde{b} = \sum_i p_i C_i b_i$, the conditional mean update is again $x - \alpha(Sx - \tilde{b})$, with the unique fixed point $\bar{x} = S^{-1}\tilde{b}$.

To verify the bias directly under the present assumptions, stationarity still gives $\sum_i p_i g_i = 0$ and

$$S(\bar{x} - x^*) = - \sum_i p_i (C_i - (1 - \varepsilon/2)I) g_i.$$

The clientwise spectral interval bounds each centered operator norm by $\varepsilon/2$. Weighted Cauchy–Schwarz and $\|S^{-1}\| \leq 1/m$ then give $\|\bar{x} - x^*\| \leq \varepsilon\zeta/(2m) = B$. Also $M = I - \alpha S$ has spectrum in $[0, \rho]$ because $0 < \alpha \leq 1/L$, and hence $\|M\| \leq \rho < 1$. Thus every deterministic ingredient needed below has been established without an independent-branch premise.

Filtration and covariance mixture. Let $\mathcal{G}_t$ be the sigma field generated by all $e_{i,k,s}$ and $e'_{i,j,k,s}$ with $s < t$, together with deterministic x_0 . The finite affine update makes x_t measurable with respect to $\mathcal{G}_t$. Induction gives a finite second moment as in the first proof: each update combines a finite number of square-integrable vectors through deterministic matrices. All current innovations are independent of $\mathcal{G}_t$ and mutually independent across the distinct underlying indices. Consequently, $\mathbb{E}[Z_{i,t} | \mathcal{G}_t] = 0$.

The common part and individual part of $Z_{i,t}$ have zero conditional cross covariance, and distinct summands within either part also have zero conditional cross covariance. Using the bound on each $R_{i,k}$ and the fact that $\|(I - \eta s_j H_i)^k\| \leq 1$ gives

$$\begin{aligned} \mathbb{E}[\|Z_{i,t}\|^2 | \mathcal{G}_t] &\leq \frac{\theta}{K^2} \sum_{k=0}^{K-1} \sigma_i^2 + \frac{1-\theta}{K^2} \sum_{j=0}^{r-1} \sum_{k=0}^{K-1} w_j^2 \sigma_i^2 \\ &= \frac{\sigma_i^2}{K} [\theta + (1-\theta)W_2]. \end{aligned}$$

Independence across clients then yields, for $Z_t = \sum_i p_i Z_{i,t}$,

$$\mathbb{E}[Z_t | \mathcal{G}_t] = 0, \quad \mathbb{E}[\|Z_t\|^2 | \mathcal{G}_t] \leq \frac{\theta + (1-\theta)W_2}{K} \sum_i p_i^2 \sigma_i^2 = V_\theta.$$

The p_i^2 factor results from client independence; it would not follow from a merely marginal bound with unspecified dependence across clients. The W_2 factor appears only in the individual part because common innovations are combined before squaring their transfer.

Finite-time recursion. The server update centered at the fixed point is $x_{t+1} - \bar{x} = M(x_t - \bar{x}) - \alpha Z_t$. Conditional expectation removes its mixed term, since $M(x_t - \bar{x})$ is $\mathcal{G}_t$ -measurable and the conditional mean of Z_t is zero. It follows that

$$\mathbb{E}[\|x_{t+1} - \bar{x}\|^2 | \mathcal{G}_t] \leq \rho^2 \|x_t - \bar{x}\|^2 + \alpha^2 V_\theta.$$

Iterating after taking expectations gives

$$\mathbb{E}\|x_t - \bar{x}\|^2 \leq \rho^{2t} \|x_0 - \bar{x}\|^2 + \alpha^2 V_\theta \frac{1 - \rho^{2t}}{1 - \rho^2}.$$

Adding $\bar{x} - x^*$, applying the squared triangle inequality with factor two, and substituting the directly proved bound B yields (19). This is exactly the claimed replacement of V in both finite-time estimates. For $\theta = 1$, substitution into V_θ removes W_2 and leaves $K^{-1} \sum_i p_i^2 \sigma_i^2$.

Samplewise full-order endpoint. If $r \geq K$, all lags $k \in \{0, \dots, K-1\}$ satisfy $k < r$, so $R_{i,k} = I$. The polynomial q_K also has degree less than r , giving $C_i = I$. If additionally $\theta = 1$, the coefficient of every individual innovation is zero. Therefore the exact unrolled identity, before taking any expectation, reduces to

$$u_{i,t} = H_i x_t - b_i + \frac{1}{K} \sum_{k=0}^{K-1} e_{i,k,t},$$

which is (20). Couple a direct minibatch process to the algorithm by using the same x_0 and the same realized $e_{i,k,t}$. Equality of the initial states and the displayed update gives equality at time one. If their states agree at time t , the same deterministic gradient and innovation average give agreement at time $t+1$. Induction proves equality of the full trajectories. This argument asserts equality under a specified coupling of processes, not only equality in distribution.

Exact scalar variance limit. For a scalar curvature $h > 0$, define $R_k(h) = \sum_j w_j (1 - \eta s_j h)^k$. When every underlying innovation has exact variance σ^2 , the same independence calculation gives equality rather than a bound:

$$\text{Var}(Z_\eta) = \frac{\sigma^2}{K^2} \left\{ \theta \sum_{k=0}^{K-1} R_k(h)^2 + (1 - \theta) \sum_j w_j^2 \sum_{k=0}^{K-1} (1 - \eta s_j h)^{2k} \right\}.$$

As $\eta \downarrow 0$, $R_k(h)$ tends to $\sum_j w_j = 1$, and every individual propagation factor tends to one. There are only finitely many terms, so taking the limit term by term gives $\sigma^2[\theta + (1 - \theta)W_2]/K$ exactly. No Gaussian premise is needed for this calculation; centering, independence, and the specified variances suffice.

Boundary checks and conclusion. At $\theta = 0$, the law reduces to independent individual innovations and the proxy becomes V ; the present proof remains valid under its additional step restriction. At $\theta = 1$, common innovations remain random unless their second moments vanish. For $K = 1$, every lag is zero and the full-order identity holds for all $r \geq 1$. For $r = 1$, $W_2 = 1$, consistent with a single branch, and the variance proxy is independent of θ . If all $\sigma_i = 0$, all innovations vanish almost surely. At $t = 0$, the geometric sum is zero, and $0 \leq \rho < 1$ keeps its denominator nonzero. The conclusions therefore include the degenerate noise and exact-interpolation endpoints without invoking any incompatible branch-independence assumption. This completes the proof of Theorem 4. $\square$